

Mathematics A

Unit 2 • Chapter OL-T15 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 19 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T15 • Expertise Q16+ • 5 items • 19 marks

Chapter	Items	Marks	Starts
01. OL-T15 Direct & Inverse Proportion	5	19	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Direct & Inverse Proportion

Unit 2 • Expertise • 19 marks • 5 item(s)

June 2025 | 4MA1-1H Q18 | 4 marks

Recover the input under inverse variation

June 2023 | 4MA1-2H Q16 | 5 marks

Root direct variation $y = k\sqrt{x}$

June 2022 | 4MA1-2H Q17 | 4 marks

Power direct variation $y = kx^n$

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

WORKING SPACE

Inverse proportionality

4 marks

Find the constant

$$y = \frac{k}{x^3}, \quad 4 = \frac{k}{3^3} \Rightarrow k = 108.$$

Use y=864

$$864 = \frac{108}{x^3} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}.$$

Final answer

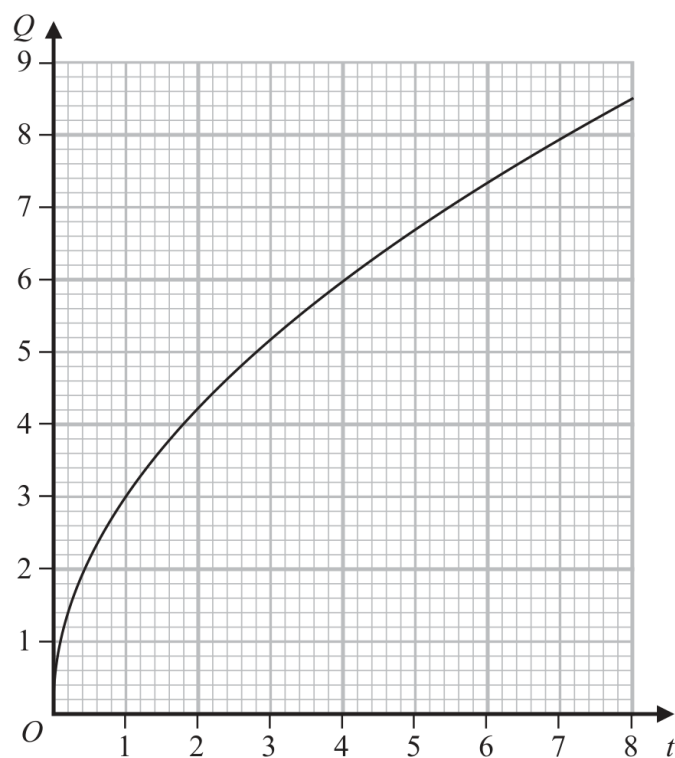
$$x = \frac{1}{2}$$

4MA1-2H Q16

5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

..... %
(2)

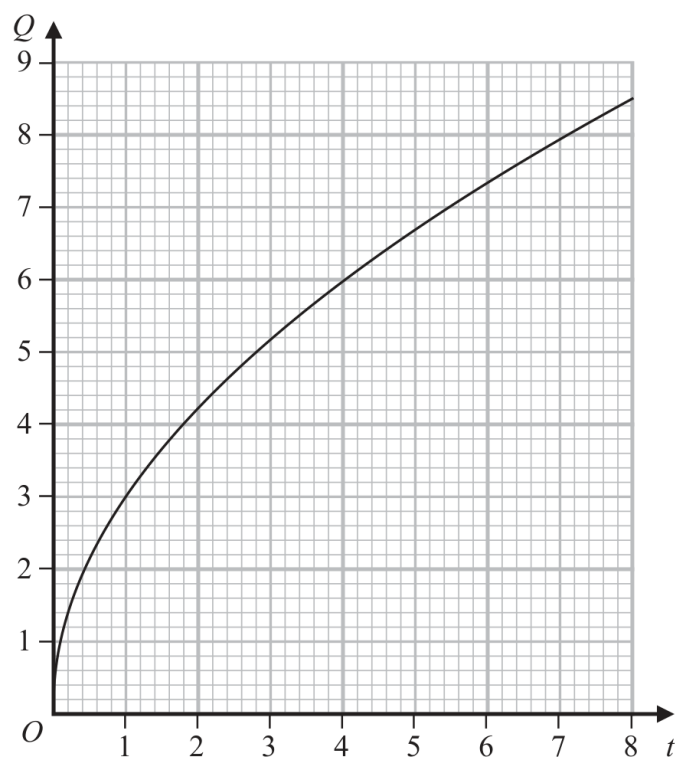
Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q16

5 marks

- 16 Q is directly proportional to $\sqrt{t}$
The graph shows the relationship between Q and t for $0 < t < 8$



- (a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

- (b) Find the percentage increase in t

..... %
(2)

Root proportionality and percentage change

5 marks

Read the graph

$$Q \propto \sqrt{t}, \quad Q = k\sqrt{t}; \text{ a point such as } (1, 3) \text{ gives } k = 3.$$

Write the model

$$Q = 3\sqrt{t}.$$

Apply the 20% change

$$Q' = 1.2Q \Rightarrow 3\sqrt{t'} = 1.2(3\sqrt{t}) \Rightarrow t' = 1.2^2 t = 1.44t.$$

Convert to a percentage

$$(1.44 - 1) \times 100\% = 44\%.$$

Final answer

$$Q = 3\sqrt{t}; \quad 44\%$$

4MA1-2H Q17 Q17

4 marks

- 17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

WORKING SPACE

4MA1-2H Q17 Q17

4 marks

- 17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

Chapter: Direct & Inverse Proportion

4 marks

Q17 – Chapter: Direct & Inverse Proportion

Family: Build an algebraic direct-variation model • Variant: Power direct variation $y=kx^n$ **Method**Direct variation gives $M = kh^3$.*Why: Translate the variation statement into an equation.***Method** $4 = k(0.5)^3$, so $k = 32$.*Why: Use the given pair to determine the constant.***Method** $500 = 32h^3$, so $h^3 = 15.625$.*Why: Substitute the new M value and isolate h cubed.***Method** $h = \text{cube root of } 15.625 = 2.5$.*Why: Take the real cube root.***Final answer****h:** $h = 2.5$

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method / working

1. Write inverse-square proportionality as $A=k/r^2$.
2. Use $A=5$ when $r=.3$: $5=k/(.3)^2$, so $k=.45$.
3. Therefore $A=.45/r^2$.

Final answer: $A=0.45/r^2$

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method / working

1. Substitute $r=7.5A$ into $A=.45/r^2$.
 2. Rearrange: $A^3=.45/(7.5^2)=.008$.
 3. Take the positive cube root: $A=.2$.
- Final answer: $A=0.2$